TAPE·FOUR

MODEL TF-4 · 4-TRACK CASSETTE MULTITRACKER

OWNER'S MANUAL

How to record with TAPE·FOUR: the deck, the cassettes, the tape switch, noise reduction, instruments and MIDI. Each control comes with what you should expect to hear when you use it.



WINDOWS · MACOS · LINUX

OCTOBER 2026

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1 WELCOME

TAPE·FOUR is a four-track cassette recorder that runs on your computer. It records onto a virtual cassette, works with any audio interface your computer has, on Windows, macOS or Linux, and plays plug-in instruments and effects (CLAP, VST3 and LV2 everywhere, Audio Units on a Mac and LADSPA on Linux) as if they were plugged into the deck.

It is deliberately not a DAW. Everything about it follows the workflow of a portable cassette multitracker:

- **Four mono tracks on one side of a tape, all running in one direction.** The tape is a C-46, C-60 or C-90, and the length of your song is limited by the length of the tape.
- **No undo.** Recording replaces whatever was on the tape under the armed tracks. To keep a take, don't record over it.
- **Everything goes to tape as audio.** There are no MIDI tracks. Each track has a signal chain in front of it, the way a rack of gear sits in front of a real deck: instruments (plug-ins, the built-ins or a hardware synth), MIDI effects and audio effects. You play it, and what comes out of the chain is recorded as sound.
- **The tape is part of the sound.** Each cassette type has its own headroom, treble limit, hiss and grain, taken from measurements of real tapes. The tape switch, noise reduction, speed and every bounce change the sound, just as they do on a real deck.

ABOUT THE SCREENSHOTS

The screenshots in this manual were taken on a test machine with no audio interface or MIDI devices connected. That's why the setup display reads `NO AUDIO DEVICE` and the audio tracks read `INPUT NOT ACTIVE`. On your computer these show your interface and its inputs.

2 GETTING STARTED

INSTALLING

Open the repository's **Actions** tab on GitHub, pick the latest green **Build** run, and download the one for your computer. Nothing needs installing. The version number (0.2.*build*) is in the window's title bar.

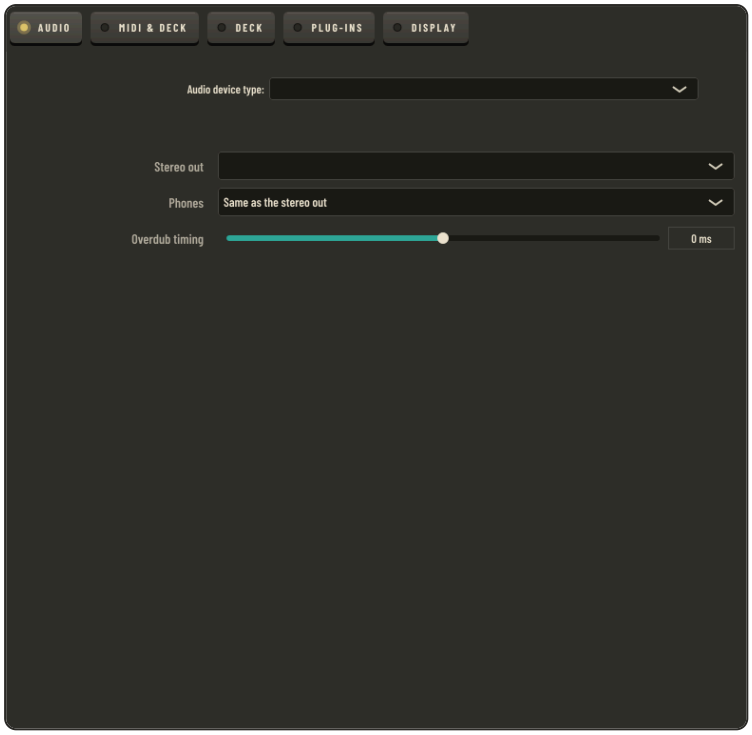
COMPUTER	DOWNLOAD	THEN
Windows, 64-bit	TapeFour-windows-x64	Unzip and run `TapeFour.exe`. Windows may warn that it comes from an unknown publisher, because it isn't code-signed: choose **More info → Run anyway**.
Mac, Apple silicon or Intel, macOS 11 or later	TapeFour-macos-universal	Unzip and move `TapeFour.app` to Applications. It isn't notarised, so the first time, right-click it and choose **Open**. Allow microphone access when asked: macOS asks this for any program that records from an audio input.
Linux, 64-bit	TapeFour-linux-x64	Extract the archive and run `TapeFour`. It uses ALSA, which every distribution has, and JACK when a JACK server is running. It plays CLAP, VST3, LV2 and LADSPA plug-ins.

TAPE·FOUR needs a 64-bit computer with at least 4 GB of memory. The loaded tape is held in memory and takes the same again on disk: 350 MB for a C-46, 460 MB for a C-60 and 690 MB for a C-90. The deck itself uses little processor time (about 1% of one core while playing); plug-ins are what cost.

The window opens where you last left it. Starting TAPE·FOUR again while it's running brings the running copy to the front.

AUDIO INTERFACE

Click the display at the top right of the deck (`SETUP ›`) to open Setup, then go to the **Audio** tab.



Setup → Audio. Choose the audio system and your interface, then the sample rate and buffer size, and the outputs.

SETTING	WHAT IT DOES	WHAT TO EXPECT
Audio device type, device	Chooses the audio system (see the table below) and the interface.	The setup display on the deck shows the system, sample rate, buffer size and round-trip latency. The first time you start, TapeFour opens the computer's usual device; on Windows it picks an ASIO driver if one is installed.
Sample rate, buffer	The interface's sample rate and block size.	A smaller buffer gives lower latency when you play live, but uses more CPU. 128 samples at 48 kHz is a good starting point.
Stereo out	The output pair the mix goes to.	The master fader controls this pair.
Phones	A second output pair, or the same as the stereo out.	The **PHONES** knob on the master strip sets its level, separately from the master fader.
Overdub timing	±50 ms trim on top of the latency the driver reports.	Use it if parts recorded through the interface (audio inputs and hardware synths) land slightly early or late against what is already on tape. Positive values move recordings earlier. Instruments and bounces don't go through the interface, so they aren't affected.

AUDIO SYSTEM	ON	WHAT TO EXPECT
ASIO	Windows	Your interface's own driver: the lowest latency. The best choice for overdubbing.
Windows Audio (WASAPI)	Windows	Works with any device, including the computer's own sound. **Exclusive Mode** and **Low Latency Mode** bring the latency down.
DirectSound	Windows	The oldest and slowest. For a device nothing else will open.

AUDIO SYSTEM	ON	WHAT TO EXPECT
CoreAudio	macOS	Every Mac interface. To use two interfaces at once, combine them into an aggregate device in Audio MIDI Setup.
ALSA	Linux	Straight to the hardware. A `hw:` device gives the lowest latency, but no other program can use the interface at the same time.
JACK	Linux	Listed when a JACK server (or PipeWire's JACK support) is running. TapeFour shares the interface with other programs.

Whatever the system, recordings line up with what you heard while playing: TapeFour shifts each recording by the interface's reported latency. Parts played through the interface are shifted by the round trip (input and output); instruments only by the output, since they're made inside the computer; bounces not at all, since they come straight off the tape.

Every change is saved straight away. If the saved driver can't be opened at start-up (for example, because another program is using it), the deck tells you, and keeps your choice for next time.

MIDI CONTROLLERS AND SYNTHS

In **Setup → MIDI & Deck**, tick the controllers you want to use and the MIDI outputs your hardware synths are on. Chapter 13 covers this in full.

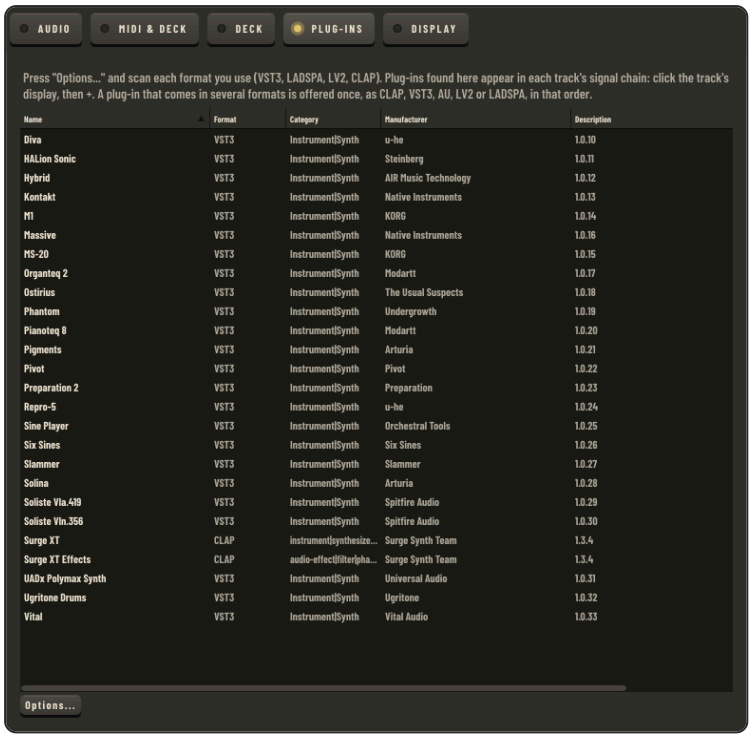
PLUG-INS

In **Setup → Plug-ins**, choose **Options...** and scan each format you use. Plug-ins found here (instruments, MIDI effects and audio effects) appear in each track's signal chain. Each plug-in is scanned in a separate helper copy of TAPE·FOUR, so a plug-in that crashes or hangs while being scanned can't take the program down: it is set aside and the scan carries on.

FORMAT	ON	LOOKED FOR IN
CLAP	Windows, macOS, Linux	Windows: `Common Files\CLAP` (in Program Files) and `%LOCALAPPDATA%\Programs\Common\CLAP`. Mac: `/Library/Audio/Plug-Ins/CLAP` and the same in your user Library. Linux: `~/.clap`, `/usr/lib/clap`, `/usr/local/lib/clap`. Also any folders in the `CLAP_PATH` environment variable.
VST3	Windows, macOS, Linux	The standard VST3 folders.
Audio Units	macOS	Whatever macOS has installed.
LV2	Windows, macOS, Linux	The standard LV2 folders (on Linux `~/.lv2`, `/usr/lib/lv2`, `/usr/local/lib/lv2`).
LADSPA	Linux	The standard LADSPA folders. LADSPA plug-ins are effects only.

Many plug-ins come in several formats. The list in a track's chain shows each plug-in once, in the best format you have: CLAP, then VST3, then Audio Units, LV2 and LADSPA. A deck you saved keeps the format it was made with. TAPE·FOUR also has three built-in instruments, **TF Transistor** (an organ), **TF**

Electric Piano and **TF Phrase Sampler**, and three built-in effects, **TF Plate**, **TF Spring Reverb** and **TF Tape Echo**, so you can start without any plug-ins.



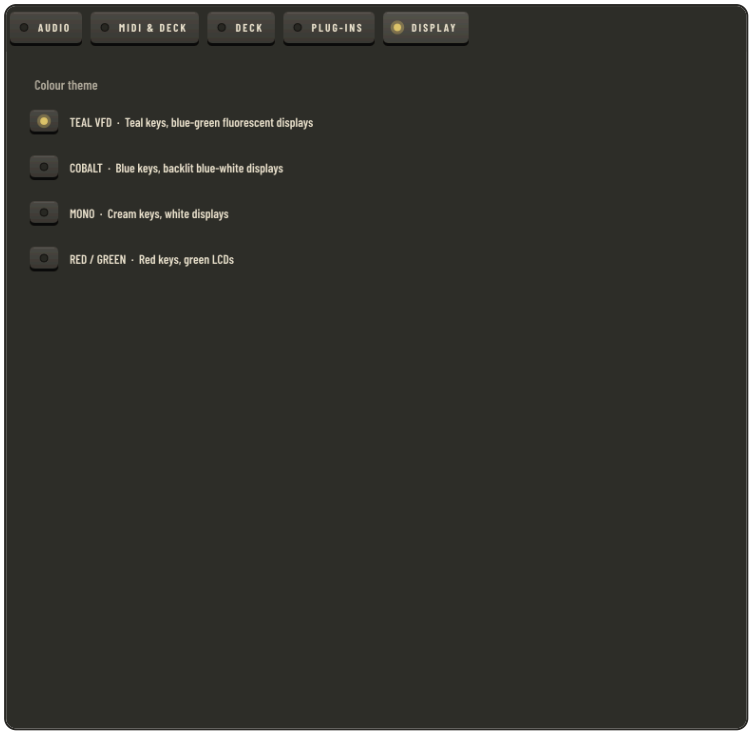
Setup → Plug-ins. Scan once; the list is remembered.

COLOURS

In **Setup → Display**, choose the deck's colour theme. It changes at once and is remembered.

THEME	KEYS AND LABELS	DISPLAYS
Teal VFD (as it comes)	Teal	Blue-green, like the fluorescent displays on 1980s decks
Cobalt	Blue	Blue-white, like a backlit LCD
Mono	Cream	White: no colour but the lamps
Red / green	Red	Green, like an unlit LCD

The theme colours the deck, not the cassettes: each cassette keeps its own print. The pictures in this manual show Teal VFD.



Setup → Display. Pick a theme; the deck changes at once.

3 THE FRONT PANEL

The deck is laid out like a portastudio: the mixer runs the full height on the left, and the tape section is a column on the right, with the cassette at the top and the transport below it. The window can be resized; the deck keeps its 16:9 shape.



Every key on the deck works the same way: the LED above its label lights when it's on, and the key itself never changes colour. The colours mean the same thing everywhere: red for record and arm, green for play and a tape that matches the switch, amber for a mode that's on or a mismatch, orange for mute and yellow for solo. Keys that just do something once (EJECT, RESET, RTZ) have no LED.

1. **Setup display.** Shows the audio driver and buffer. Click it to open Setup.
2. **MIDI IN / MIDI OUT lamps.** They flash when notes come in from a controller or go out to a hardware synth.
3. **Cassette.** The tape moves from the left spool to the right as it plays. Double-click the label to write on it.
4. **Inlay.** Click it to take the J-card out and write the track list.
5. **EJECT.** Puts in a new blank tape or loads one from the shelf. It fades while the tape is running: stop the tape first.
6. **Tape switch.** Three keys, NORMAL, CHROME and METAL bias and EQ; the lit one is where the switch is (chapter 6).
7. **Tape counter**, with **RESET** (set 000 here) and **RTZ** (return to 000).
8. **Status display.** Transport state, time remaining, and warnings such as mismatches.
9. **Transport keys**: REC, REW, F.FWD, PLAY, STOP. The LEDs in REC and PLAY light while they're engaged.
10. **Pitch**: varispeed, smoothly from −12% to +12%.
11. **Tape speed**: 1⅞ or 3¾ inches per second; the lit key is the speed.
12. **Noise reduction** on or off; its LED shows which.
13. **NR type display.** Click it to choose the type, and the rest of the deck, in Setup.
14. **Track display.** What plays the track, which controller and input feed it, its signal chain as a row of blocks, and the NR, EQ and MIDI lamps. Click it to open the track's signal chain (chapter 8); double-click it to open the window of the first plug-in in the chain.
15. **Gain, EQ and pan.** HIGH, MID with its FREQ knob, and LOW.
16. **Meter and fader.** 0 VU on the meter is Dolby level on tape.
17. **MUTE, SOLO and ARM**, each with its own lamp. Mute silences the track, solo picks what you hear, and an armed track plays its live input and records when you press REC (chapter 9).

18 **Master strip.** Phones level, stereo meter with its peak readout, and master fader.

19 **LEARN.** MIDI learn: map knobs, faders and buttons on your controller to the deck (chapter 13).

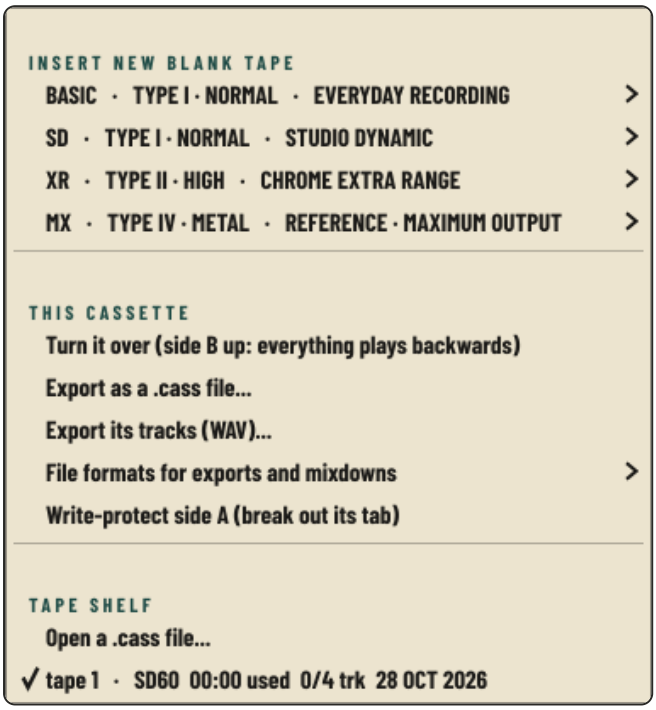
20 **Master deck display.** The master tape, the side and how much is on it, and the file format. Click it to open the master deck (chapter 14).

21 **SIDE and MIXDOWN.** Turn the master tape over, and arm a mixdown: PLAY then records the mix (chapter 14).

4 CASSETTES

PUTTING IN A TAPE

With the tape stopped, press **EJECT**. The menu's top part inserts a new blank tape: choose the cassette (BASIC, SD, XR or MX), then the length, then its condition. The middle part works on the cassette in the deck. The bottom part, the **tape shelf**, lists every tape you have made, and the tape that's in the deck is ticked.



The eject menu. New blank tapes, the cassette in the deck, and the shelf.



The tape shelf. Each entry shows the label, cassette, minutes used, tracks used and date.

LENGTH	PER SIDE	AT 1⅞ IPS	AT 3¾ IPS	SPOOLS
C-46	23 min	23:00	11:30	Smallest packs
C-60	30 min	30:00	15:00	Medium
C-90	45 min	45:00	22:30	Largest packs

Like a real 4-track, TAPE·FOUR records all four tracks across the full width of the tape, running the same way (you can turn the cassette over, below). When the tape reaches the end, the transport stops and the status display shows `END OF TAPE`.

A BLANK TAPE'S CONDITION

A cassette's condition belongs to the cassette, as on a real one, so you choose it when you put in a blank. It is saved with the tape and travels with it.

CONDITION	HISS	TREBLE	MOD. NOISE	DROPOUTS
NEW	as measured	as measured	as measured	almost none
USED	+0.5 dB	−0.5 dB	+1 dB	one every few minutes
WORN	+1.5 dB	−1.5 dB	+3 dB	about one a minute
CHEWED	+3 dB	−4 dB	+6 dB	several a minute

Dropouts sit at fixed places on the tape, so a dropout comes back at the same moment every time you play that stretch. A tape also wears and ages slowly as you use it (chapter 12). Tapes made before this version are given the condition closest to their old settings.

TURNING THE TAPE OVER: REVERSE RECORDING

EJECT → TURN IT OVER puts the cassette in side B up, as if you had taken it out and turned it round. The header above the cassette turns amber and the label shows **B**. Like a real 4-track, the deck now sees the tape from the other end:

- everything on it plays **backwards**;
- the tracks come in **mirrored**: what you recorded on track 1 is under the track 4 head, and track 2 is under track 3.

That is the classic way to get sounds that only exist backwards. Turn the tape over, record a cymbal, a guitar line or a reverb tail on a free track, and turn the tape back: it now plays in reverse, swelling into the beat.

Side A's start is side B's end, so after turning over you will usually rewind first. The counter keeps the number it showed, as on a real deck. Noise reduction flags, the tape switch, dropouts and wear stay with their places on the tape; decoding NR on a recording that plays backwards breathes, as it does on a real deck. The deck remembers which side is up along with the tape. **EXPORT ITS TRACKS** always exports side A, so turn the tape back first.

WRITE-PROTECT TABS

EJECT → WRITE-PROTECT SIDE A (or **B**, when side B is up) breaks out the tab of the side facing up and stops the deck recording on that side, as on a real cassette: pressing **REC** shows `WRITE-PROTECTED` instead. Choose it again to tape over the hole.

CASSETTE FILES (.CASS)

EJECT → EXPORT AS A .CASS FILE saves the cassette in the deck as a single file you can keep, send to someone or open on another computer. It holds the tape and nothing else: the four tracks (16-bit, uncompressed), what was recorded where and at what speed, the formulation, condition and wear, the label and the tabs. Your mixer settings and plug-ins stay behind. A C-60 is about 460 MB, a C-90 about 690 MB.

EJECT → OPEN A .CASS FILE, or opening a `.cass` file with TAPE·FOUR, puts a copy on your shelf and loads it. The format is open and documented with the source code (`docs/formats`), so other programs can read and write it.

EXPORTING THE TRACKS

EJECT → EXPORT ITS TRACKS writes each track with something on it to its own file, as it plays back off the tape on this deck (speed, noise reduction, the tape switch and the tape's condition included). Every file starts at the start of the tape, so they line up with each other. **EJECT → FILE FORMATS** chooses the format:

FORMAT	CHOICES	USE IT FOR
WAV (default)	16-bit, 24-bit, 32-bit float	Taking the tracks into another program at full quality.
FLAC	16-bit, 24-bit	The same quality in about half the space.
MP3	128 to 320 kbps	Sending a listen to someone.

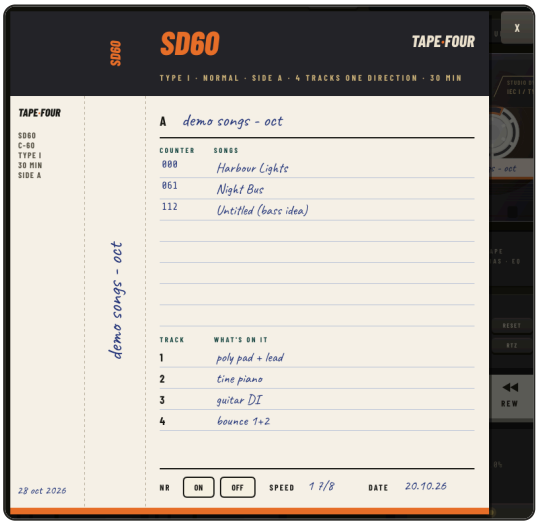
Tick several, or **ALL THREE**, to get every format at once. Mixdowns use the same choice.

SAVING

You never need to save. A tape is written to disk once it has been stopped for a second, and again whenever you change it. It is also saved when you eject it and when you quit. The deck remembers which tape was in when you quit and loads it next time.

LABEL AND INLAY

Double-click the label on the cassette to write on it. Click the inlay beside the cassette to take the J-card out of its case and write on it: the title, up to eight songs with their counter numbers, a note for each track, whether you recorded with NR, the speed, and the date. Click outside the card, press `Esc` or click **X** to put it back. The title on the inlay and the label on the cassette are the same text, so changing one changes the other.



The inlay, opened out: back flap, spine and front. Its colours follow the cassette type.

5 THE FOUR TAPE TYPES

Each cassette in TAPE·FOUR behaves like a real tape. Its headroom, treble limit, hiss, grain and output level come from published lab measurements of a real cassette of that kind.

The figures are in dB relative to **Dolby level**, the tape's reference level. On TAPE·FOUR that is **0 VU** on the meters, or −12 dBFS. All figures are at 1⅞ ips without noise reduction.

TERM	MEANING	WHAT YOU HEAR
MOL 400 Hz	Maximum output level: where a 400 Hz tone reaches 3% distortion.	How loud you can record before the tape thickens and then distorts. Bass gets a little less room than the mids, as on real tape.
SOL 10 kHz	Saturation output level for treble.	How loud cymbals, hi-hats and bright synths can go before the tape squashes them. Treble saturates long before the mids, especially at 1⅞ ips.
S/N	Hiss on blank tape, A-weighted, below Dolby level.	The hiss in quiet passages.
Mod. noise	Modulation noise: grain that rides on the signal.	A fine roughness on sustained notes. It gets louder and quieter with the music, so noise reduction can't remove it.
Sensitivity	How loud the tape plays back what you recorded.	A small level shift on every pass through the tape. You notice it after several bounces.
EQ	The playback EQ time constant: 120 µs or 70 µs.	70 µs tapes have less treble hiss.



BASIC

TYPE I · NORMAL BIAS · 120 µs · MODELLED ON SONY HF

A budget ferric tape, the kind that came in multipacks. It has the least headroom and the most hiss and grain of the four. It sounds warm and slightly fuzzy, and compresses early if you record hot. It's good for sketches, lo-fi textures and anything you want to sound like an old demo.

MOL 400 / 100 Hz	+2.4 / +2.0 dB
SOL 10 kHz	−10.3 dB
S/N	54.3 dB(A)
Mod. noise	−36.1 dB
Sensitivity	−1.0 dB



SD

TYPE I · NORMAL BIAS · 120 µs · MODELLED ON TDK AD

A premium ferric tape and the all-rounder. It has good mid-range headroom, the best treble headroom of the Type I tapes, and moderate hiss. Tapes made before tape types existed play as SD. It's the tape to reach for when you don't have a reason to pick another.

MOL 400 / 100 Hz	+6.2 / +3.8 dB
SOL 10 kHz	−8.2 dB
S/N	60.3 dB(A)
Mod. noise	−44.3 dB
Sensitivity	−0.3 dB



XR

TYPE II · HIGH (CHROME) BIAS · 70 µs · MODELLED ON TDK SA

A chrome-position tape. Its 70 µs playback EQ makes the hiss quieter and less bright, and it plays back slightly hot. It has the least treble headroom of the four, so loud bright sounds dull and compress first, especially at 1⅞ ips. It's clean and quiet for vocals, acoustic instruments and pads. For bright, loud material, use 3¾ ips or MX.

MOL 400 / 100 Hz	+4.4 / +3.4 dB
SOL 10 kHz	−11.2 dB
S/N	60.9 dB(A)
Mod. noise	−47.2 dB
Sensitivity	+1.1 dB



MX

TYPE IV · METAL BIAS · 70 µs · MODELLED ON MAXELL MX

A metal tape. It has the most headroom everywhere (about 3 dB more mid-range and 1.4 dB more treble than SD), the lowest grain and the lowest hiss. It takes hot levels and bright material without complaint. It's the closest a cassette gets to transparent.

MOL 400 / 100 Hz	+9.1 / +8.0 dB
SOL 10 kHz	−6.8 dB
S/N	62.7 dB(A)
Mod. noise	−50.4 dB
Sensitivity	+0.8 dB

CHOOSING A TAPE: WHAT TO EXPECT

IF YOU…	BASIC	SD	XR	MX
record a vocal peaking at +3 VU	Audibly thick, a little gritty	Clean, slightly rounded	Clean	Clean
record hi-hats at 0 VU at 1⅞	Softened, a bit dull	Mostly intact	Noticeably dulled	Intact

IF YOU…	BASIC	SD	XR	MX
listen to blank tape (NR off)	Most hiss	About 6 dB less	Less, and darker	Least
hold a long synth note	Slight roughness	Faint	Very faint	Practically none
bounce three times	About 3 dB quieter	About 1 dB quieter	About 3 dB louder	About 2.4 dB louder

Generation loss comes from each tape's sensitivity, which is applied on every pass. Noise reduction doubles it (chapter 11).

6 THE TAPE SWITCH

The **TAPE · BIAS · EQ** switch under the cassette sets three things at once, as on a real deck: the record bias, the record EQ and the playback EQ.



A mismatch. An MX (metal) tape is loaded but the switch is at NORMAL. The LED on the NORMAL key is amber, and the status display reads `BIAS LOW: TYPE IV TAPE AT NORMAL`.

- **When a tape goes in, the switch moves to that tape's type.** After that it stays where you put it, so a mismatch is always your choice. Restarting the audio device doesn't move it.
- **One key is lit: where the switch is.** Its LED is green when that position suits the loaded cassette and amber when it doesn't. The status display then says which way the bias is off.
- **Every stretch of tape remembers the switch position it was recorded at**, just as it remembers noise reduction and speed.

BIAS: RECORDING AT THE WRONG POSITION

A Type II tape needs about 1.5 times the bias of a Type I, and a Type IV about 2.5 times.

- **Too little bias (under-bias)** leaves the tape bright and fuzzy. It distorts and compresses sooner and gets grainier.
- **Too much bias (over-bias)** wipes out the treble as it records. The tape sounds dull and a little quieter, but slightly cleaner.

These effects are recorded into the tape. The table assumes you play the tape back at the same position you recorded at.

TAPE ↓ · SWITCH →	NORMAL	CHROME	METAL
Type I: BASIC, SD	Correct.	Over-biased by 3.5 dB. About −7 dB at 10 kHz, −0.9 dB overall, slightly cleaner (+0.9 dB MOL). Muffled.	Over-biased by 8 dB. About −16 dB at 10 kHz and −2 dB overall. Very dull, like a worn head.
Type II: XR	Under-biased by 3.5 dB. About +2.6 dB at 10 kHz, MOL −5.3 dB, grain +4 dB. Bright and edgy; distorts early.	Correct.	Over-biased by 4.5 dB. About −9 dB at 10 kHz and −1.1 dB overall. Dull.
Type IV: MX	Under-biased by 8 dB. MOL −12 dB, grain +10 dB, −2.5 dB overall, treble about −1 dB. Heavily distorted and gritty. **Old recordings bleed through** (see below).	Under-biased by 4.5 dB. About +2.5 dB at 10 kHz, MOL −6.8 dB, grain +5 dB. Bright and crunchy.	Correct.

ABOUT THESE FIGURES

The bias ratios between tape types follow the tape standards. How much the treble, headroom and grain move per dB of bias error is an estimate of typical real-deck behaviour, not a published measurement. The automated tests check every row of this table.

PLAYBACK EQ: PLAYING BACK AT THE OTHER POSITION

NORMAL plays back with 120 µs EQ, while CHROME and METAL both use 70 µs. Each stretch of tape is played back through the EQ of the switch as it is now, compared with the EQ it was recorded with:

RECORDED AT	PLAYED AT	RESULT
CHROME or METAL (70 µs)	NORMAL (120 µs)	Treble lifted by a shelf starting near 1.3 kHz: about +4.5 dB at 8 kHz, up to +4.7 dB. Bright and hissy.
NORMAL (120 µs)	CHROME or METAL (70 µs)	The reverse: about −4.5 dB at 8 kHz. Muffled.
CHROME	METAL, or the other way round	No EQ change; both use 70 µs.

Blank-tape hiss follows the same rule: a Type II or IV tape played at NORMAL hisses brighter. You can combine both effects. For example, record a chrome tape at NORMAL (under-biased and bright), then play it back at CHROME.



The EQ lamp. These tracks were recorded at NORMAL and the switch is now at CHROME, so each track's EQ lamp is red and the lit switch key is amber. The EQ lamp flashes while the tape plays. It is green when the recording's EQ matches the switch, and off on blank tape.

METAL TAPE AT NORMAL: OLD MATERIAL BLEEDS THROUGH

On a NORMAL setting the erase head isn't strong enough to fully erase high-coercivity metal tape. If you record over an MX tape at NORMAL, about 30 dB of the old recording survives underneath the new one, mostly its bass. Treble erases more easily. Record over it at METAL and the old recording is fully gone (it measures below −90 dB).

TRY THIS

Record a bass line on an MX tape at METAL. Set the switch to NORMAL and record a new part over it on the same track. Play it back: the old bass line is still there underneath, faint and muffled, like a ghost of the earlier take.

7 TRANSPORT AND COUNTER

CONTROL	WHAT IT DOES	WHAT TO EXPECT
PLAY `Space`	Plays the tape.	Unarmed tracks play the tape; armed tracks keep playing their live input, so you can rehearse over the tape. The LED in the PLAY key lights.
STOP `Space`	Stops.	One second later the tape is saved.
REC `R`	Records on every armed track.	The LED in the REC key lights. If no track is armed, the status display says `ARM A TRACK FIRST`.
REC while playing	Punch-in.	Recording starts at once on the armed tracks. Press PLAY to punch out and keep playing. Tape before and after the punch is untouched. The new take fades in and out over 6 ms either side, like the gap between a deck's erase and record heads, so a punch never clicks.
REW, F.FWD, pressed once	Cue (forward) and review (back).	The tape winds at about a third of full speed with the head touching it lightly. You hear it faintly, high and fast going forward and backwards in review, so you can find a song by ear: the gaps between songs stand out as short silences. The status display reads `CUE` or `REVIEW`. Press PLAY to play from there. Pressing the other winding key reverses, still listening. REC does nothing while cueing.
REW, F.FWD, pressed again	Wind at full speed.	The head lifts clear and the tape winds silently. The motor drives the take-up spool, so the tape starts slowly and speeds up as that spool fills: about 9 times playing speed at the start of a C-60 side and 20 times at the end. A whole C-60 side takes about two minutes. Winding stops when the leader runs out at either end.
RESET	Sets the counter to 000.	Use it to mark the start of a song.
RTZ `Home`	Return to zero.	Rewinds until the counter reads 000, and stops. The status display reads `RETURN TO ZERO`.

THE COUNTER

The counter belongs to the deck, not to the cassette, as on a real machine:

- **It counts turns of the right-hand spool, not seconds.** At the start of a side that spool is small and turns quickly, so the counter runs about twice as fast as at the end. A song's counter number is only true on this deck, which is why you write it on the inlay.
- **It keeps counting when you change the cassette or turn it over.** Only RESET puts it back to 000. It also keeps its reading when you quit.
- **You can switch to a real-time counter** (minutes and seconds) in Setup → DECK. The deck's counter type is part of the machine and is saved in `TF-4.deck`.

The status display shows the time left on the side at the current speed, or `LEADER` while the leader is across the head.

LEADERS

Each end of a cassette is a few seconds of clear leader. It has no oxide, so nothing records on it and it doesn't hiss. A new cassette comes rewound, with the leader across the head; press PLAY or REC and the magnetic tape arrives a few seconds later, as on a real cassette. Every cassette's leaders are a little different, and stay the same wherever it's played.

NO UNDO

Recording replaces what was on the tape under the armed tracks, from the moment you press REC until you stop or punch out. Nothing else is touched. To keep a take, record on a different track or a new tape.

8 SIGNAL CHAINS

Every track has a signal chain in front of the tape, like a rack of gear wired up before a real deck: up to eight blocks in a row. MIDI from your controller runs along the top of the chain and audio along the bottom, and whatever comes out of the last block is what the track records.

Click a track's display to drop down its chain. Click a block to select it, and its settings appear underneath. Click outside, click **X** or press `Esc` to close it.

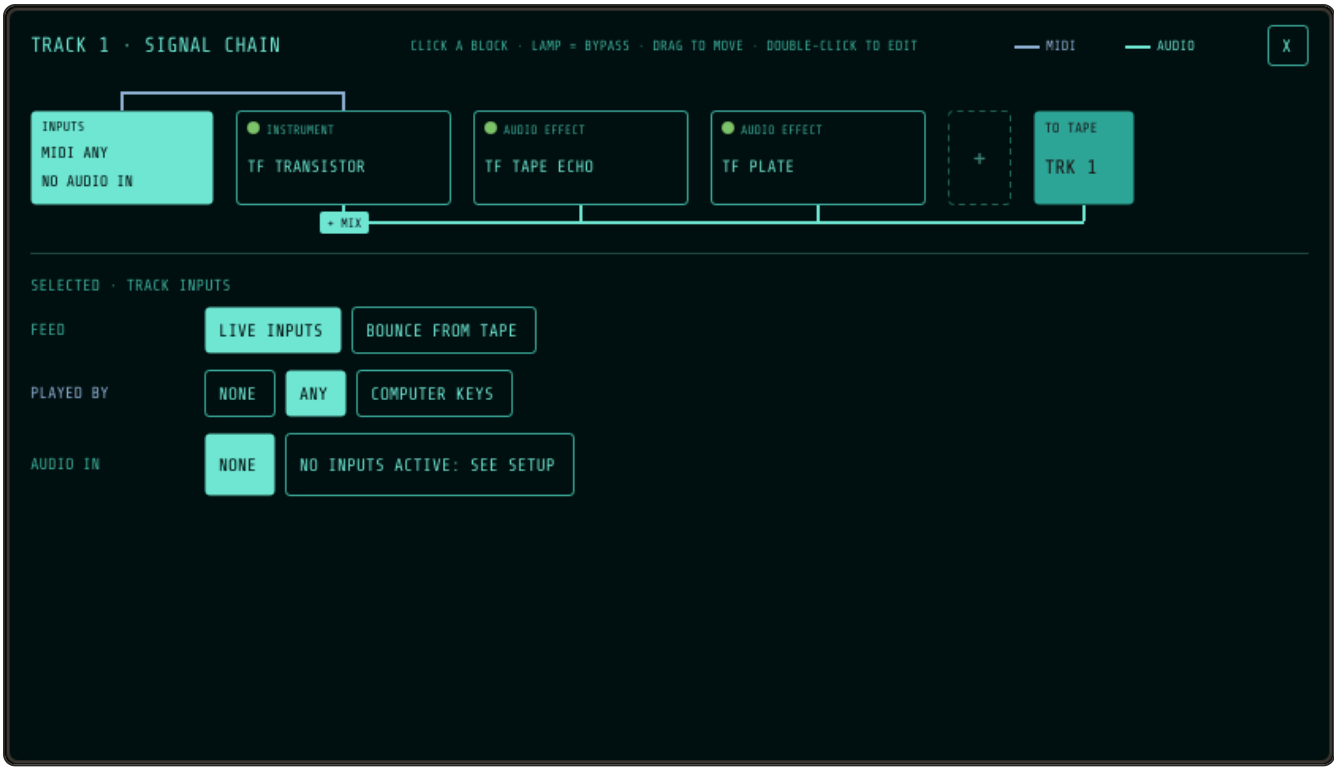


Track 1's chain. The inputs, then TF Transistor, TF Tape Echo and TF Plate, then the tape. The blue line is MIDI: it stops at the organ, the last block that listens to it. The amber line is audio: it starts where the organ mixes in (`+ MIX`). TF Plate is selected; underneath are its controls and the list to replace it from.

TO	DO THIS
Add a block	Click **+** at the end of the chain and pick from the list.
Bypass a block	Click its lamp, or **BYPASS**. A bypassed block is dimmed and the signal passes it untouched.
Reorder	Drag a block left or right. A bar shows where it will land.
Open a plug-in's window	Double-click the block, or **EDIT**. The window stays on top, and `Space`, `R` and `Home` still work the transport from it.
Replace a block	Select it and pick another from the list beside it. The new block takes its place in the chain.
Remove a block	**REMOVE**, `Delete`, or right-click it.

THE INPUTS

The first block is the track's inputs: what plays it and what it hears.



The inputs. On the test machine there is no interface, so no audio inputs are listed; on yours, each input appears with a level bar so you can see which one your instrument is plugged into.

SETTING	CHOICES	WHAT TO EXPECT
Feed	**LIVE INPUTS** or **BOUNCE FROM TAPE**	Live: the controller and audio input below feed the chain. Bounce: other tracks are mixed off the tape into the chain instead (see Bouncing, below).
Played by	**NONE**, **ANY**, or a named controller	Which controller's notes go into the chain. **Any**: every ticked controller. **A named controller** plays only this track, so different keyboards can play different armed tracks at once. Notes only reach the chain while the track is armed.
Audio in	**NONE**, **IN N**, or **IN N+M**	An input of your interface, or a pair mixed to mono. Tracks are mono; a stereo source on a pair is summed at half level per side. A track can have both a controller and an audio input: play a synth and sing through the same chain.

THE BLOCKS

There is one kind of chain, not separate instrument and effect slots. Each block does what it is:

BLOCK	WHAT IT DOES WITH MIDI	WHAT IT DOES WITH AUDIO
Instrument	Plays the notes that reach it.	Mixes its sound in with whatever is already in the chain (`+ MIX`), at its **LEVEL**. Two instruments one after the other layer, like two synths into a mixer. An instrument with an audio input (a vocoder, say) can instead **PROCESS INPUT**.
MIDI effect	Changes the notes for every block after it: an arpeggiator, a chord generator, or a retuner such as PitchGrid for MPE instruments. The lane after it is drawn brighter and tagged `CHANGES MIDI`.	Passes it through.

BLOCK	WHAT IT DOES WITH MIDI	WHAT IT DOES WITH AUDIO
Audio effect	Passes it through.	Processes the audio, blended with the dry signal by its **MIX**.
External instrument	Sends the notes out to a hardware synth.	Mixes in the synth's audio from the input it's plugged into.

Order matters, as it does with real gear. A MIDI effect only changes the blocks after it. An effect placed before an instrument doesn't hear that instrument, only what came before it. So a typical chain is *MIDI effects → instruments → audio effects*, but nothing stops you putting a reverb on the voice coming in, then a synth on top, then a delay over both.

Every instrument comes in at its own level: plug-ins vary a lot, so each instrument block has its own **LEVEL**, from −24 to +12 dB. The strip under each instrument's node in the chain, and the segments under its **LEVEL**, show what it is sending into the track, on the same scale as the track meters. Set **LEVEL** so your loudest playing peaks around 0, then set the recording level with the track's gain and fader as usual. The built-in instruments are set up so ordinary playing peaks around 0 as they come.

A track is mono, and the chain is stereo: at its end the two sides are added together with equal power. A stereo instrument keeps its level however wide its sound; one that's the same on both sides comes in 3 dB higher. An input comes through at the level it went in.

PICKING A BLOCK



Adding a block. The built-ins first, then hardware synths, then every scanned plug-in, grouped by maker.

The tabs narrow the list to **INSTRUMENTS**, **MIDI FX** or **AUDIO FX**. Type to search; `Return` takes the first match. Two plug-ins with the same name are both listed, with their folder and file so you can tell them apart. **FIND PLUG-INS…** takes you to Setup → Plug-ins to scan. Plug-ins load in the background; until one arrives its block reads `LOADING`, and the chain keeps its order.

BUILT-IN	KIND	WHAT TO EXPECT
TF Transistor	Instrument	A transistor combo organ after the 1960s Vox Continental, built as its circuit is: twelve oscillators divided down into every octave, four contacts a key, drawbars and vibrato. It has its own editor (below).
TF Electric Piano	Instrument	A reed electric piano after the Wurlitzer 200 and 200A, built from its mechanism: reeds, felt hammers, the charged pickup that makes it bark, dampers, the preamp's vibrato and the little speakers. It has its own editor (below).
TF Phrase Sampler	Instrument	A phrase sampler after the 1995 Yamaha SU10: 48 pads of samples in its small memory, as grainy as the original's, played from the mouse or MIDI, with the ribbon's filter, pitch, scratch and crossfade, and four songs. It samples the track's input, the tape or a file. It has its own window (below).
TF Plate	Audio effect	A smooth, dense plate reverb with pre-delay. Size, damping and pre-delay.
TF Spring Reverb	Audio effect	The boingy, dripping reverb of a guitar amp's spring tank, physically modelled: three real tanks, and the valve that drives them. It has its own window (below).
TF Tape Echo	Audio effect	A tape delay whose repeats darken and wobble as they fade. Time, feedback, tone and wow.

THE ELECTRIC PIANO

Select **TF Electric Piano** in a chain and press **EDIT** (or double-click it): its editor opens in a window of its own, like a plug-in's, so you can move it aside and keep working on the mixer. It has eight tabs, the way a technician goes through the instrument. Every control can be learned (**LEARN**) and is saved with the deck. It plays 15 notes at once, in mono, as the real piano does: place it with the track's pan.



MODEL. A 200A, serviced: what the piano is like now, in one box.

TAB	WHAT IT HOLDS
MODEL	**200** or **200A**: the 200's amp is noisier and its Alnico speakers smoother; the 200A's middle and upper reeds are a little thicker. A condition (**NEW**, **SERVICED**, **WORN**, **NEGLECTED**) sets the technician's controls for a piano in that state; change any of them afterwards. With **AGES AS YOU PLAY** on, the piano wears as you play it: over hundreds of hours its reeds drift out of tune, its hammer felt hardens, its damper felt softens and reeds start to ring short. Choosing a condition starts its playing time again.
TUNING	Equal temperament, just intonation, Pythagorean, quarter-comma meantone, Werckmeister III, Kirnberger III and Vallotti (on any key); **PER NOTE**, where you drag each of the 12 notes up or down in cents; **EDO**, any number of equal steps to the octave from 5 to 72 (type it, or step it); or a Scala scale (`.scl`) with its keyboard map (`.kbm`). A4 from 415 to 466 Hz, and octave stretch. The reeds' own drift comes on top.
REEDS	How far the reeds have drifted from the tuning, how long they ring, and how many are short ringers. The chart shows all 64 (A1 to C7): how long each rings, and how far out of tune it is. Short ringers are lit.
HAMMERS	Felt hardness, strike point, blow distance, let-off and the velocity curve. A hammer stays on its reed for about one cycle of the note, so it strikes mostly the fundamental. Harder felt and a longer blow make a shorter, harder hit, and the reed swings further. Too big a let-off and the softest notes don't reach the reed at all.
PICKUP	Each reed sits between two fingers of a charged comb, its face level with theirs and resting just clear of them (a paper's thickness, as a technician sets it). **REED AT REST** moves it; rested inside the fingers, a note comes out harsh, as the service manual warns. **SIDE GAP** is the clearance either side; narrower is louder. The polarising voltage, 160 to 170 V on a healthy piano, scales the output. Played softly, a reed swings a little way into the fingers and back: a round tone with its octave. Harder, it fills them and the tone thickens. Hardest, it swings right through and the tone folds over: the bark. The bass, whose reeds swing furthest, barks first; the top stays clean. The trace shows all three.
DAMPERS	How firmly the dampers stop a note, and the noise of the dampers, the action and the pedal. The sustain pedal is your controller's.
AMP	Volume, vibrato depth (at the 200A's fixed 5.75 Hz), preamp drive, mains hum (50 or 60 Hz), and the output: **SPEAKERS**, the power amp and the two small oval speakers in the lid as the player hears them, or **REAR JACK**, the preamp's own output.
PLAYING	MPE is off as it comes: the piano hears only how hard you play, like the real one. Turned on, it can follow each note's pitch bend, and pressure can raise its volume, vibrato, drive or damp the note under your finger; slide can move the hammer felt or the reed height.



TUNING. Werckmeister III in C: each note's offset from equal temperament.



PICKUP. Middle C's reed between its fingers, end on, and what the pickup sends played softly, medium and hard.

PRESETS

The built-in instruments keep presets the same way; the piano's are described here. A preset keeps the whole piano: every control, its tuning (a Scala file and keyboard map included), its reeds as they drifted, and how long it has been played. The box at the top left of the editor's window shows the preset it came from, with * once you've changed a control (playing it, and the wear that comes with that, doesn't count). Click the box for your presets, in place of the tabs:

- **SAVE** writes the piano over its preset (with none, it asks for a name). **SAVE AS NEW** asks for a name and keeps it as another. Return saves, Esc cancels.
- **RENAME** and **DELETE** act on the current preset; deleting asks once.
- Click a preset to load it. What you had is replaced, so save it first if you want it.
- **SHOW FOLDER** opens `Documents/TapeFour/Presets/Electric Piano`. Each preset is one file there, so you can copy them to another computer or share them.



Presets. Naming a new one; the current preset is lit.

THE TRANSISTOR ORGAN

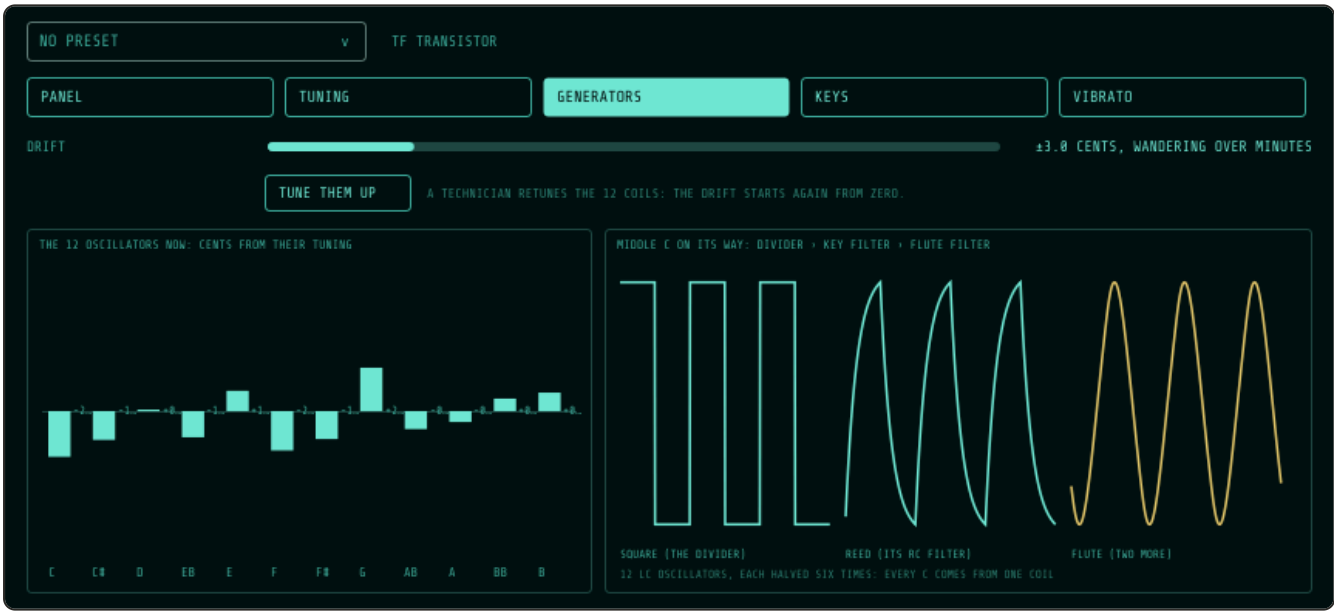
Select **TF Transistor** in a chain and press **EDIT** (or double-click it). Its editor opens in a window of its own, like the piano's: first the organ's own panel, then what's inside its case. Like the original, it plays any number of notes at once, with no velocity, out of one output.

Inside, twelve oscillators make the twelve notes of the top octave. Each feeds a chain of dividers that halve it again and again, so every C on the organ comes from one oscillator, and every octave is exact. Each key closes four contacts, one each for 16', 8', 4' and the IV mixture, switching those pitches straight on.



PANEL. The six drawbars and the vibrato switch, and what middle C sounds pitch by pitch (amber: the flute tone's share).

TAB	WHAT IT HOLDS
PANEL	The drawbars: drag one down to bring it in, double-click it for its default. **16'**, **8'** and **4'** set those pitches; **IV** is the mixture, four higher pitches together (2⅔', 2', 1⅗' and 1'). The two amber drawbars set the tone: **~** the soft flute tone, **M** the bright reed tone. With both in, the organ is silent. **VIBRATO** on or off.
TUNING	Equal temperament, just intonation, Pythagorean, quarter-comma meantone, Werckmeister III, Kirnberger III and Vallotti (on any key); **PER NOTE**, where you drag each of the twelve oscillators up or down; or a Scala scale of 12 notes to the octave. A4 from 415 to 466 Hz. With only twelve oscillators and exact octaves, an organ like this can't have the piano's EDO or octave stretch.
GENERATORS	The oscillators drift, as germanium transistors do: **DRIFT** sets how far they wander, and the chart shows where each is now. **TUNE THEM UP** is the technician retuning them. Alongside: middle C's square wave from its divider, rounded by its filter into the reed tone, and smoothed into the flute tone.
KEYS	**CLICK**: the contacts switch instantly, as on the original, or you can soften them. **BOUNCE**: how long a contact chatters as it closes and opens. **CONTACT SPREAD**: how far apart a key's four contacts close, so the pitches don't all start together. The diagram shows middle C's four contacts as the key goes down and comes up.
VIBRATO	Speed and depth, which on the original are set inside the case. One wobble moves all twelve oscillators, so every note moves together.



GENERATORS. The twelve oscillators' drift, and middle C on its way from divider to flute tone.



KEYS. Middle C's four contacts: bouncing as they close, a few milliseconds apart, and again as the key comes up.

Presets work as the piano's do (above), in `Documents/TapeFour/Presets/Transistor`.

THE PHRASE SAMPLER

TF Phrase Sampler is modelled on the Yamaha SU10, a battery-powered sampler from 1995 with twelve pads and a ribbon. Select it in a chain and press **EDIT** (or double-click it); its window opens on its front panel. It's an instrument that takes the track's audio, so put it on the track whose input you'll sample from (it's added with `PROCESS` on).

All its samples share one small memory, 384 KB, as on the original. How long that lasts depends on the *grade*, the sampler's one sample rate: **HI** sounds best and holds about 19 seconds in all, **STD** 27, **LONG** 38 and **EXLG** 54 (half that in stereo). The grade is for every sample at once: change it later and every sample plays faster and higher, or slower and lower. Samples are stored in four bits, which gives them the original's gritty, slightly crunchy sound.



PLAY. The front panel. FILTER is on, and the drum loop on pad 3 is playing.

PLAYING IT

- **Pads.** Click a pad to play its sample; click lower on the pad to play louder. **BANK A** to **BANK D** switch between four sets of 12, so there are 48 in all. From MIDI, bank A's pad 1 is C1 (note 36) and bank D's pad 12 is B4; velocity sets the level.
- **The ribbon** is the long strip. Drag along it to work the effect that's lit: **PITCH** bends every sample from −20 % to +10 %; **FILTER** sweeps a filter (its type and resonance are on SETUP); **SCALE** plays one sample across all 12 pads in semitones (press it, then the pad with the sample); **CROSS FADER** fades from the samples to the track's input. Press the lit key again to turn it off.
- **The input.** With no effect on, you hear only the samples, not the track's input. **EXT FILTER** puts the input through the ribbon's filter (pads 1–4 then choose low, band, high pass or none, and 7–9 the resonance); **EXT SCRATCH** plays the input, and touching the ribbon freezes it so you can scratch what just went past.
- **SCRATCH.** Hold **SCRATCH** and press a pad: that sample now lies under the ribbon like a record under your hand. Drag to scratch; **BWD**, **FWD** and **TOP** move along it. Press **SCRATCH** again to leave it.
- **HOLD** keeps pads sounding after you let go; **REV**, held, plays them backwards.
- **How many at once.** Four samples at most (a stereo one counts as two), fewer at HI and with some effects; the lights under VOICES show what's free. Another cuts the oldest. At HI the ribbon's effects and SCRATCH aren't available, and at STD neither are FILTER and SCALE, as on the original.

SAMPLING

On **SAMPLE**, choose the source: **TRACK INPUT** (what comes into this track), one of the four tracks as the tape plays it, or a file (WAV, AIFF, FLAC, Ogg or MP3). Choose the grade, stereo or mono, and the pad it goes to (it must be empty). Set the **GAIN** so the INPUT meter reaches the amber but not the red. **START**: **MANUAL** records as soon as you press **RECORD**; a level waits for a sound that loud. Press **STOP** when you have it;

recording also stops when the memory is full. A file goes in the same way, as if it had been played into the input, and is cut short if it doesn't fit.

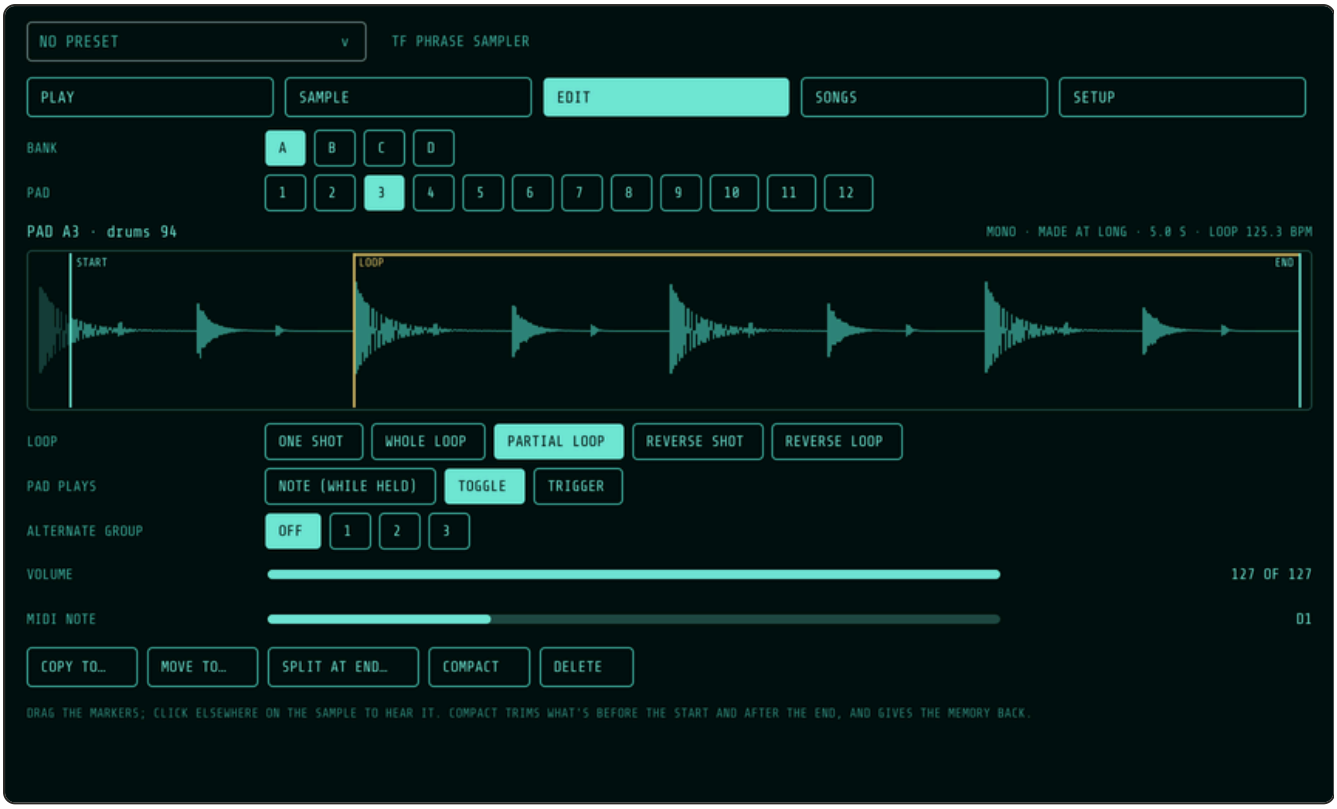


SAMPLE. The memory bar shows each pad's share; the pad being edited is amber.

EDITING A PAD

On **EDIT**, choose a pad (the last one you played is chosen for you). Drag `START` and `END` to trim it; click elsewhere on the picture to hear it. **LOOP** sets how it plays: once, looping the whole thing, looping from the `LOOP` point (drag it), or the same backwards. **PAD PLAYS**: **NOTE** sounds while held, **TOGGLE** starts on one press and stops on the next, **TRIGGER** plays to the end however briefly you press. Pads in the same **ALTERNATE GROUP** cut each other off, like an open and a closed hi-hat. For a loop, the header shows its tempo.

Trimming doesn't free any memory until you press **COMPACT**. **COPY TO…**, **MOVE TO…** and **SPLIT AT END…** (the part after the end point becomes another pad) ask for an empty pad. **DELETE** asks you to click twice.



EDIT. A drum loop trimmed at the start, looping from its loop point.

SONGS

The sampler records what you play on the pads, in time, as one of four songs. On **SONGS**, press **RECORD** beside an empty song; it starts with the first pad you play (on the PLAY tab or from MIDI) and ends at **STOP**. **PLAY** plays it back; with HOLD on, it repeats. On the front panel, a **SONG** key chooses the song for **REC** and **PLAY**. A song can't be added to: delete it and record it again. The ribbon isn't recorded, and the four songs share about 1000 presses and releases.



SONGS.

SETUP

PITCH tunes every sample at once (it changes the sample rate, recording included); **VOLUME** is the output level. The filter's type and resonance are here too. **MIDI CHANNEL** is ANY or one channel. The ribbon can follow your controller's pitch bend or a controller number (to begin with CC 1, the mod wheel): let go and it lets go. **PADS ON C1–B4** puts every pad back on its own note.



SETUP.

A preset keeps everything: every sample, the pads, the songs and the settings. Presets work as the piano's do (above), in `Documents/TapeFour/Presets/Phrase Sampler`. The deck keeps them too, so the sampler's sounds are still there when you open the deck again.

THE SPRING REVERB

TF Spring Reverb is a spring reverb tank, the kind inside guitar amplifiers: a sound drives one end of a few long springs, and a pickup at the other end hears what arrives. Select it in a chain and press **EDIT** (or double-click it) for its window. Its **MIX** is on the chain, as for the other effects.

Sound travels along a spring at different speeds for different pitches, so a click arrives as a "chirp", and comes back again and again as it bounces between the ends. That is the drip and splash of spring reverb. Here the springs are worked out from their physics, so the chirps, echoes and decay come from the springs themselves.



TF Spring Reverb. The Fender tank, from above, and a click through it: each curve is a chirp, and the chirps keep coming as the sound goes back and forth along the springs.

CONTROL	WHAT IT DOES
TANK	Three real tanks: **2 SPRING LONG**, the one in Fender amps, with the longest decay; **3 SPRING**, denser; **3 SPRING SHORT**, a small tank, tighter and quicker.
MODEL	**SIMULATED SPRINGS** works out each spring's vibration; **WAVEGUIDE** is a simpler model of the same springs, a little brighter.
DWELL	How hard the valve drives the springs. Low settings are clean; towards 10 the valve clips and the springs crash and splash, the surf sound.
TONE	Darker to the left, brighter to the right.

The window's picture shows what a click turns into, pitch upwards and time across. Like a real tank, it plays in mono.

HARDWARE SYNTHS



An external instrument, on a USB MIDI interface.

Tick the synth's MIDI output in Setup → MIDI & Deck (you can name it there), then add it from **Hardware synths** in the list. Choose where its MIDI goes and on which channel (**AS PLAYED** keeps an MPE controller's notes on their own channels), and which input its audio comes back on. Changing the route sends all-notes-off so nothing hangs; **ALL NOTES OFF** does it by hand. Its audio is recorded in time with the tape, allowing for the round trip through the interface.

BOUNCING

Set a track's feed to **BOUNCE FROM TAPE** and tick the tracks to mix onto it. While recording, you hear the sources once, through their own EQ, pan and fader; use those to balance the bounce, then arm the destination and record. A muted source is left out. The bounce still goes through the track's chain, so you can put a reverb over a whole submix as you bounce it. It picks up another pass of the tape: a little more hiss and grain, and a level shift of the tape's sensitivity.

MISSING PLUG-INS

If a chain uses a plug-in this computer doesn't have (a tape made on another machine, or a plug-in that was removed), its block reads `MISSING` and is skipped. Its settings are kept: rescan in Setup → Plug-ins and it comes back as it was, or pick a replacement.

READING THE TRACK DISPLAY



Track displays. Line 1: what plays the track. Line 2: its controller and audio input. Line 3: the chain, as blocks (dim ones are bypassed). Line 4: what the track is doing, and three lamps.

The tag above the display says what feeds the track: `MIDI`, `AUDIO`, `AUDIO+MIDI` or `BOUNCE`. Line 4 reads `LIVE IN` when the track is armed (you hear its inputs, not the tape), `MUTED` when it's muted, and `→ TAPE` otherwise. When the tape under the head was recorded at a different speed from the one you're playing at, it shows the pitch shift in semitones instead, for example `-12.0 SEMITONES`.

LAMP	LIT	RED
NR	The tape under the head was recorded with noise reduction, or you're recording with NR on.	Flashes red while playing if the NR switch doesn't match how the tape was recorded.
EQ	Green: the tape here was recorded with the same playback EQ as the switch.	The switch's EQ differs from the recording's (chapter 6). It flashes while playing.
MIDI	Green flash: notes are reaching this track.	—

9 THE MIXER

Each of the four channel strips has its controls in a single column, top to bottom. Double-click any knob or fader to reset it: gain to 0 dB, EQ and pan to centre, FREQ to 1.2 kHz, fader to the 0 mark.

CONTROL	RANGE	WHAT TO EXPECT
GAIN	−20 to +40 dB (0 dB at the default)	Input trim before the tape. Use it to put peaks near 0 VU. Higher levels reach the tape's MOL and compress or distort (chapter 5).
HIGH	±12 dB shelf: about ±11.6 dB at 10 kHz, ±7 dB at 5 kHz, half way at 4.5 kHz	Presence and air, well inside what the tape can record. Full boost is clearly brighter, and also brings up hiss.
MID	±12 dB peak	Presence, honk or boxiness, depending on FREQ. A cut makes a part sit back.
FREQ	250 Hz to 5 kHz (1.2 kHz at the centre)	Where MID works; the smaller knob with the blue pointer. Around 250–500 Hz for body or mud, 1–2 kHz for honk, 3–5 kHz for bite and vocal presence. Sweep it with MID boosted to find a spot, then cut or boost.
LOW	±12 dB shelf: about ±11.6 dB at 100 Hz, ±6 dB at 250 Hz	Weight. A cut helps when bouncing several tracks together.
PAN	L100 to R100, constant power	Centre is −3 dB to each side. Hard left puts the full level on the left output only.
Fader	−∞ to +7 dB (0 at the mark)	The level into the mix and into bounces. On an armed track it also sets the level going to tape, as on a portastudio.
Meter	20 LEDs reading peaks, 0 VU at −12 dBFS	Green below 0 VU, amber from 0 to +3, red above +3. The highest peak stays lit for a moment. It reads peaks, not loudness: a sound with sharp attacks reads higher than a dense one that sounds as loud. Keep peaks around 0 VU for a clean recording, and push into the amber for tape colour.
MUTE	On or off (orange lamp)	Silences the track everywhere: in the mix, in bounces, and its live input. An armed track that's muted **still records to tape**: you just don't hear it.
SOLO	On or off (yellow lamp)	Changes only what you hear. With any track soloed, only soloed tracks are heard. Solo doesn't change what is recorded or bounced.
ARM	On or off (red lamp)	An armed track plays its live inputs through its chain instead of its tape, whatever the transport is doing, and the display reads `LIVE IN`. The lamp flashes while armed and stays lit while recording. Press REC on the transport and it records to tape. To hear a take, disarm the track: while it is armed you hear the input, not the tape.

Mute wins over solo. A track that is both soloed and muted stays silent. If every soloed track is also muted, nothing is heard, and the status display says `SOLOED TRACK IS MUTED` so you know why.

The **master strip** has a stereo meter, the master fader for the stereo out, and the **PHONES** knob, which sets the level of the phones output pair independently.

WHAT THE EQ AND GAIN RECORD, AND WHAT THEY DON'T

On a cassette 4-track the channel strip sits between the input and the tape. TAPE·FOUR works the same way: gain, EQ and the fader shape what is recorded on an armed track. On playback, the same strip shapes what you hear from the tape. To fix a recording later, adjust the strip on playback, or bounce it through the strip onto another track.

10 SPEED AND PITCH

SETTING	WHAT IT DOES	WHAT TO EXPECT
1⅞ ips	Normal cassette speed.	The tape's full length. Treble saturates sooner and there's slightly more hiss.
3¾ ips	Double speed, as on many 4-track portastudios.	Half the recording time. About an octave more treble headroom (the treble limit moves up), and about 3 dB less hiss in the audible range.
Playing at the other speed	Every stretch of tape remembers the speed it was recorded at.	A recording made at 3¾ and played at 1⅞ sounds an octave lower at half speed, and the reverse sounds an octave higher. The track display shows the shift in semitones. Record at 1⅞ and play at 3¾ for chipmunk vocals; do the reverse for slow, dark textures.
PITCH	Varispeed, smoothly from −12% to +12% (about ±2 semitones). Double-click to return to 0.	Speeds up or slows down the tape and changes pitch and tempo together. The tape glides to a new setting over a fraction of a second, like a capstan motor, so turning it while the tape plays bends the pitch rather than jumping. Use it to tune the tape to an instrument, or to record a part slowed down and play it back at normal speed. The status display shows the setting, for example `PLAY` $1\frac{7}{8}$ `+3.7%`.

Wow, flutter and drift (chapter 12) are added on top of the pitch setting, so the tape always moves slightly unevenly unless you turn them off.

11 NOISE REDUCTION

Noise reduction encodes what you record and decodes on playback. On the way to tape it boosts quiet sounds, mostly the treble. On the way back it turns them down again, and the tape hiss goes down with them. Turn it on with the **NOISE REDUCTION** button. Choose the type by clicking the **NR TYPE** display, or in **Setup → DECK**.

TYPE	HOW IT WORKS	HISS REDUCTION	CHARACTER
2:1 compander	As dbx type II, the system on most portastudios: the whole signal is compressed 2:1 on record and expanded 1:2 on playback, steered by a detector that listens mostly to the treble, with fixed treble emphasis.	The most, 30 dB and more	Very quiet between notes. Because the whole band moves, the hiss rises and falls with the music ("breathing"), most noticeably on sparse, bright parts.
Sliding band B	As Dolby B: quiet treble is lifted on record, starting about 500 Hz and reaching 10 dB from about 4 kHz up. As a sound gets louder the lifted band slides up above it, so loud sounds are left alone and the treble above them stays quiet.	About 10 dB, in the treble	Gentle and forgiving; little breathing.
Sliding band C	As Dolby C: two such stages, one working 20 dB below the other, reaching two octaves lower. Quiet sounds get up to 20 dB of lift (about 15 dB at 400 Hz). Very high treble is trimmed before recording and restored after, and loud treble is held back from the tape so it doesn't saturate.	About 20 dB, from the mid-range up	Quieter than B, and more sensitive to level and treble errors: the most demanding of a good deck and tape.

The three types follow the published designs: B matches Dolby Laboratories' own encoder response, and C the Dolby C test points, within a fraction of a dB. The tape's hiss is generated on the tape and passes through the decoder like the music, so you hear exactly what each system does to it.

MISMATCHED NR

Tape remembers whether it was recorded with NR. If you play it back with the switch the other way, the track's NR lamp flashes red. Encoded tape played without NR sounds bright and squashed, and the treble pumps. Plain tape played with NR sounds dull and lifeless.

HOW THE TAPE TYPE AND NOISE REDUCTION INTERACT

Noise reduction expects the tape to give back exactly what the encoder put on it. Anything that changes level or treble on the way through gets enlarged by the decoder, not just passed along. In TAPE·FOUR the encoder sits before the tape model and the decoder after it, so all of this happens on its own:

- **Hiss floor.** NR removes a fixed amount, so the tape's own hiss still matters. MX with C or 2:1 is the quietest combination, and BASIC with B the noisiest.
- **Level errors grow.** With 2:1, a 1 dB difference becomes 2 dB. SD (−0.3 dB) loses about 0.6 dB per pass with NR on. XR (+1.1 dB) gains about 2.2 dB. B and C mistrack in the treble on quiet passages.
- **Treble errors grow and move.** A bias or EQ mismatch is a treble error. B and C read it as a level change, so a bright mismatch gets brighter on quiet passages and a dull one duller. The treble

"breathes" with the music. With 2:1 the treble error is doubled throughout.

- **Treble saturation.** The encoder lifts quiet treble before the tape. On tapes with little treble headroom (XR and BASIC, especially at 1⅞ ips), loud bright parts saturate after encoding. The decoder then mistracks, and cymbals turn dull and swishy. MX has the most room.
- **Grain.** NR can't remove modulation noise, which rides on the signal. A 2:1 compander's gain swings make it, and the hiss, rise and fall. That's most obvious on BASIC.

12 THE DECK AND THE TAPE'S CONDITION

As in the real world, some of the sound belongs to the machine and some to the cassette. The machine's wow, flutter, heads and crosstalk stay the same whatever tape is in it. The cassette's formulation, condition and wear go with the cassette, wherever it is played.

THE DECK: SETUP → DECK



Setup → DECK, with a new SD60 loaded.

CONTROL	RANGE	WHAT TO EXPECT
Wow	0–0.6%	A slow pitch waver (under 1.5 Hz). It's most audible on sustained notes and piano.
Flutter	0–0.3%	A fast waver (6–12 Hz). It gives a slight shimmer, and above 0.15% a watery roughness.
Speed drift	0 to ±1%	The tape speed wanders slowly, every few seconds, as with tired motor belts.
Heads	Treble to 4–20 kHz	Worn or misaligned heads lose treble on playback, whatever the tape. It dulls everything played back, hiss included.
Crosstalk	None, or −100 to −40 dB	Neighbouring tracks leak into each other. You may hear it as a faint ghost of track 2 behind track 1.
Tape drive	−6 to +6 dB (centre = 0 VU at Dolby level)	Where 0 VU lands on the tape. More drive brings saturation and treble compression earlier, the way a deck calibrated hot would. Less drive keeps it cleaner. The playback level isn't changed. Double-click to return to the centre.
Mixer noise	−100 to −60 dB per channel (as supplied: −76 dB)	The mixer's own hiss, as on any analogue desk. Each channel adds its own, which follows its fader and pan; the mix bus adds a little of its own that doesn't. You rarely hear it over the tape, but with noise reduction on, between notes, it's what remains. The default is TASCAM's figure for the 424.

CONTROL	RANGE	WHAT TO EXPECT
Fast wind	60–200 s per C-60 side (as supplied: 120 s)	How long the machine takes to wind one side end to end.
Noise reduction	2:1, B or C	See chapter 11.
Counter	Mechanical or real time	See chapter 7. Changing it resets the counter.

The machine presets set the mechanism and heads (the first five controls). The mixer's noise and the wind time belong to the machine too, but don't change with its condition:

MACHINE	WOW	FLUTTER	DRIFT	HEADS	CROSSTALK
NEW	0.08%	0.045%	±0.1%	18 kHz	−70 dB
SERVICED (as supplied)	0.15%	0.075%	±0.2%	15.2 kHz	−65 dB
WORN	0.30%	0.15%	±0.4%	9.3 kHz	−55 dB
NEGLECTED	0.45%	0.24%	±0.7%	6.6 kHz	−45 dB

The panel below the controls shows the cassette in the deck: its measured figures and its condition. Tick **Play every cassette as new** to have the deck ignore cassettes' condition and age; their formulation still counts. The deck's settings are saved in `TF-4.deck`, beside the settings file.

WEAR AND AGE

Real tapes change slowly, and so do these. Nothing here is exaggerated: a tape you use for months still sounds like the same tape.

- **Use.** Every pass over the heads wears that stretch of tape a little, and only that stretch (cueing counts as a quarter of a pass). After about 400 passes it has lost half a dB of treble, gained about a dB of modulation noise, and has twice the dropouts it started with. The wear slows down from there.
- **Time.** The oxide loses up to 0.3 dB of treble over the years.
- **Print-through.** A recording very faintly magnetises the turns of tape either side of it on the reel. You hear it as a quiet pre-echo and post-echo in the silence just before and after loud sounds: about 60 dB down a few days after recording, and about 50 dB down after twenty years.

13 MIDI, MIDI LEARN AND THE DECK STATE



Setup → MIDI & Deck. Controllers in, hardware synths out, MIDI control with the list of mappings, and deck recall.

The tick boxes in Setup are small LED keys too: lit is on.

CONTROLLERS IN

- **Use**: TAPE·FOUR only listens to controllers that are ticked. Each track then chooses its controller in its chain's inputs (chapter 8).
- **MPE**: tick it for MPE controllers (Osmose, Seaboard, LinnStrument and others). Known MPE controllers start ticked. A track played by an MPE controller tells its instrument about the MPE zone (lower zone, 15 channels). Each note then gets its own pitch bend, pressure and timbre (CC74). The built-in instruments respond directly. Plug-ins need MPE support, and some need it switched on in their own settings. CLAP plug-ins that take MIDI get MPE as it is played.
- **Bend range**: type the per-note pitch-bend range your controller sends (the Osmose default is 48 semitones). If it's wrong, bent notes land at the wrong pitch.
- **Computer keyboard**: `A W S E D F T G Y H U J K O L P ;` play notes like a piano keyboard, and `Z` and `X` change octave.

PLUGGING IN WHILE RUNNING

TAPE·FOUR notices MIDI devices being plugged in and out while it runs. Controllers and synth outputs you have ticked reconnect on their own, and tracks routed to them carry on. A controller TAPE·FOUR has never seen comes in switched on; untick it in Setup if you don't want it.

HARDWARE SYNTHS OUT

Tick the MIDI outputs your synths are on. You can give each one a name, such as "JUNO-106", which the chain and the track display then show instead of the port name. Add the synth to a track's chain as an external instrument (chapter 8). The MIDI OUT lamp at the top of the deck flashes when notes go out.

MIDI LEARN

Any knob, fader or button on your controller can move a control on the deck or a parameter of a plug-in.



Learn mode. Every control that can be learned is outlined: amber if it's free, green with a badge if it's already mapped (here, the KeyLab's CC 74 is on track 1's fader and a pad is on its mute).

1. Press **LEARN** at the top of the deck. Its lamp lights and the controls that can be learned are outlined.
2. Click a control. It lights in the deck's accent colour and reads `MOVE A CONTROL`.
3. Move a knob or fader, or press a button, on your controller. The status display confirms it, for example `LEARNED: TRK 1 FADER ← CC 74`.
4. Click another control to learn the next, or press **LEARN** or `Esc` to finish.

Without entering learn mode, **right-click** any of these controls and choose **Learn MIDI control** or **Forget MIDI control**.

CAN BE LEARNED	CONTROLS
Each track	Gain, High, Mid, Freq, Low, Pan, Fader, Mute, Solo, Arm
The deck	Master fader, Phones, Pitch, Tape speed, Noise reduction
Transport	Play, Stop, Rec, Rew, F.Fwd, RTZ
Plug-ins	Any parameter of any block in a chain

To learn a plug-in parameter, open the track's chain, select the block and press **LEARN** there. Either touch the control in the plug-in's own window or pick it from the list, then move a control on your keyboard. The block's mappings are listed under it, each with an **X** to forget it.



Learning a plug-in parameter. Touching Feedback in the echo's window, or clicking it here, picks it.

HOW MAPPED CONTROLS BEHAVE

- **Pickup.** A knob or fader only takes over once it passes the control's current position, so nothing jumps when you first touch it. Until then the status display tells you which way to turn it. Pickup can be turned off per mapping in the list.
- **Endless encoders** are recognised while learning (the common relative formats) and nudge the control up and down, with no pickup needed.
- **14-bit controllers** that send a CC pair (0–31 with 32–63) are combined for fine resolution.
- **Buttons and pads** learned onto a switch (mute, solo, arm, tape speed, noise reduction) toggle it each press, and onto a transport key press that key.
- **A mapped control is kept for its job.** Its messages don't also reach the track's instruments, so a knob mapped to a fader doesn't sweep a synth's filter at the same time. Tick **Also plays** in the list to pass them on as well.

MIDI MACHINE CONTROL

Many keyboards and control surfaces have transport buttons that send MIDI Machine Control (MMC). With **Follow transport buttons that send MIDI Machine Control** ticked (the default), they play, stop, record, wind and return the tape with no learning needed. A button that sends a plain CC or note can be learned onto a transport key instead.

THE MAPPINGS LIST

Setup → MIDI & Deck lists every mapping: which controller and message it's from, what it controls, how it moves (knob, 14-bit, encoder or press), and its **Pickup** and **Also plays** settings. ✕ forgets one. Mappings to the deck are kept with your settings. Mappings to a plug-in's parameters are kept with that block, so they travel with the tape and come back when the deck is recalled.

RECALL THE DECK WITH EACH TAPE

Every tape remembers the deck it was last used with:

- each track's inputs and signal chain, with every plug-in's settings and MIDI mappings;
- MIDI routing;
- mute and solo;
- gain, EQ, pan and faders;
- master and phones;
- speed and pitch;
- the tape switch and noise reduction.

A plug-in's sound is saved when you close its window, a second after the tape stops, when the tape is saved or ejected, and when you quit. While a plug-in's window stays open it's also saved once a minute while the tape is stopped, so a crash costs at most a minute of tweaking. Big plug-ins are never asked for their sound while the tape is running.

With **Recall the deck with each tape** ticked, loading a tape puts the deck back exactly as it was. Unticked (the default), the deck stays as it is when you change tapes. The tape switch always moves to the new tape's type either way.

14 MIXDOWN AND THE MASTER DECK

When the mix is right, record it to a second cassette deck: the **master deck**, a good stereo hi-fi deck. The mix goes onto a stereo master tape and is played straight back off it, as on a three-head deck, and that is what is written to the files. You hear the master bus as usual while it records.



The master deck, opened from the display on the master strip.

SETTING UP THE MASTER DECK

Click the display at the top of the **MASTER** strip.

Tape	BASIC, SD, XR or MX, as on the TF-4. **DIRECT** skips the tape and writes the mix exactly as it is.
Length	C-46, C-60 or C-90 for new master tapes. A master tape has two sides, like any cassette.
NR	Record and play the master with noise reduction, of the same type as the TF-4's.
Condition	The condition of new master tapes, as in chapter 4.
Rec level	The master deck's record level, −12 to +6 dB. It can be changed during a mix.
Files	WAV, FLAC, MP3 or all three, with their bit depth or bit rate. This is the same choice as for track exports.

Changing the tape, length or condition while the master tape is still blank swaps it for a new one of that kind. Once it has a mix on it, **NEW MASTER TAPE** puts in a fresh one, and **MASTER TAPES** loads an older one. Every master deck setting is remembered. The master deck's own machine settings are in `TF-M.deck`, beside the settings file.

MAKING A MIX

1. Press **MIXDOWN** under the master fader. Its LED lights and the status display says which mix is next.
2. Press **PLAY**, from wherever you want the mix to start. The master deck records.

3. Press **STOP** to end the mix. The master deck lets the last moment come off the tape, then finishes the files.

The PEAK readout above the master meter holds the loudest level of the mix, and the display warns if the mix clipped. Each mix goes after the last one on the current side, with a four-second gap. When a side is full the mix ends there: press **SIDE** (or click the other side in the master deck) to turn the tape over.

YOUR MIXES

The files go to `Documents/TapeFour/Mixdowns/`, in a folder named after the tape, one file per format; **SHOW FOLDER** opens it. The master deck lists every mix on the master tape and where it sits on each side. Click one to listen to it, off the master tape. **SAVE SIDE AS ONE FILE** writes a whole side, gaps and all, as one file, and **EXPORT .CASS** saves the master tape itself as a cassette file (chapter 4), rewound to the start of side A, so whoever plays it first starts at the beginning. That's the file to put on your phone for the TF-P1 Pocket player, which has its own guide.

15 REFERENCE

KEYBOARD

KEY	ACTION
`Space`	Play / stop
`R`	Record (punch in while playing)
`Home`	Return to zero
`Esc`	Close the signal chain or inlay; leave learn mode
`Delete`	Remove the selected block from a chain
`A` … `;`	Play notes; `Z` / `X` change octave

WHERE THINGS ARE KEPT

Tapes	`Documents/TapeFour/Tape Shelf/`: one folder per tape, holding the cassette (the same files as in a `.cass`: `cassette.json`, `state.json`, and each track's audio and map) and `session.xml` (the mixer and plug-ins that go with it). Tapes from earlier versions are converted the first time they're loaded, and the old files are kept with `.bak` on the end. Tapes are saved in the background: every five seconds while recording, so a crash costs seconds rather than a take; a second after the transport stops; and on eject and quit.
Mixdowns	`Documents/TapeFour/Mixdowns/`, one folder per tape.
Settings	`TapeFour.settings`: audio device, MIDI, deck MIDI mappings, the knobs and the master deck. Beside it, `TF-4.deck` and `TF-M.deck` hold the two machines. In `%APPDATA%\TapeFour` on Windows, `~/Library/Application Support/TapeFour` on a Mac, `~/.config/TapeFour` on Linux. Only written when something has changed.
Crash reports	`Crashes`, next to the settings. If TapeFour ever crashes it writes a report there (the version and a stack trace, and on Windows a `.dmp` file) and says so when it next starts.

TROUBLESHOOTING

PROBLEM	TRY THIS
NO AUDIO DEVICE	Open Setup → Audio and choose your device. If another program holds it (common with ASIO and exclusive-mode WASAPI), close that program, then reselect the device. On a Mac, check TapeFour is allowed the microphone in System Settings → Privacy & Security.
REC does nothing	Arm at least one track. The status display says `ARM A TRACK FIRST`.
A take seems to have recorded silence	While a track is armed you hear its live input, not the tape. Disarm it and play the take back.
An instrument doesn't play	Tick the controller in Setup → MIDI & Deck. Check **PLAYED BY** in the track's chain inputs, and arm the track: instruments only sound while armed. Check the instrument isn't bypassed or after a MIDI effect that filters notes out. Watch the MIDI IN lamp and the track's MIDI lamp.

PROBLEM	TRY THIS
Recordings sound distorted	Lower GAIN so peaks sit around 0 VU, use a tape with more headroom (MX), centre Tape drive, and check the tape switch matches the tape.
Everything sounds dull	Check the EQ lamps (playback EQ mismatch), the NR lamps (NR mismatch), the tape switch (over-bias) the heads in Setup → DECK, and the cassette's condition.
Bounces get quieter or louder	That's the tape's sensitivity, doubled with 2:1 NR. Correct it with the faders before each bounce.
A plug-in doesn't appear after a scan	Scan its format: each format has its own **Scan for new or updated … plug-ins** in Setup → Plug-ins → Options…. On Linux, LV2 plug-ins must be in an LV2 folder (one that holds `.lv2` bundles), and CLAP plug-ins in a CLAP folder or one named in `CLAP_PATH`.
A plug-in shows as failed after a scan	It crashed or hung while being scanned, so TAPE·FOUR set it aside. Update the plug-in, select it in Setup → Plug-ins, choose **Options… → Remove selected plug-in from list**, and scan again.
A controller plugged in later doesn't play	Check it's ticked in Setup → MIDI & Deck, that the track's **PLAYED BY** names it or is set to **Any**, and that the track is armed.
A plug-in is missing when a tape is recalled	Its block reads `MISSING`. Rescan in Setup → Plug-ins. The plug-in's settings are kept until it can be loaded again.
Nothing is heard and the status says `SOLOED TRACK IS MUTED`	Every soloed track is also muted, and mute wins. Unmute it or clear the solos.
A mapped knob does nothing at first	That's pickup: turn it past the control's current position (the status display says which way), or untick Pickup for it in Setup → MIDI & Deck.
A knob moves a control and a synth at once	Untick **Also plays** for that mapping.
`NOT ENOUGH DISK SPACE`	A blank tape is only put in if the disk can hold the whole of it (all four tracks recorded end to end). Free some space, or choose a shorter tape.
`DISK NEARLY FULL`	Shown when you press REC and the disk couldn't hold the rest of the tape. The take still records; anything that can't be written is retried at every save, so free some space before you stop.
`LAST SESSION CRASHED`	The report is in the Crashes folder (see above). The tape was saved up to five seconds before the crash.
A ghost of an old take is audible	You recorded over a metal tape at NORMAL. Record over it again at METAL to erase it fully.

CASSETTE MEASUREMENTS

CASSETTE	TYPE	MEASURED ON	MOL 400	MOL 100	SOL 10K	S/N (A)	MOD. NOISE	SENS.	EQ
BASIC	I	Sony HF	+2.4	+2.0	−10.3	54.3	−36.1	−1.0	120 µs
SD	I	TDK AD	+6.2	+3.8	−8.2	60.3	−44.3	−0.3	120 µs
XR	II	TDK SA	+4.4	+3.4	−11.2	60.9	−47.2	+1.1	70 µs

CASSETTE	TYPE	MEASURED ON	MOL 400	MOL 100	SOL 10K	S/N (A)	MOD. NOISE	SENS.	EQ
MX	IV	Maxell MX	+9.1	+8.0	−6.8	62.7	−50.4	+0.8	70 µs

dB relative to 400 Hz Dolby level (0 VU), at 1⅞ ips, without noise reduction. TAPE·FOUR's automated tests record tones through each cassette and check that these figures come out within about 1 dB, and that the tape switch's EQ and bias effects match chapter 6.

SOURCES

- Compact Cassette tape types and formulations, Wikipedia (including Roberson's measurements of retail cassettes): en.wikipedia.org/wiki/Compact_Cassette_tape_types_and_formulations
- Adaptive biasing, Wikipedia: en.wikipedia.org/wiki/Adaptive_biasing
- "Analog Tape 101, part 3: Bias Magic", Mix: mixonline.com/recording/analog-tape-101-part-3-bias-magic-373029
- TDK D, AD, SA and MA product pages, vintagecassettes.com

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